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GitHub: https://github.com/jngakwa/seattle-birds-classification

Sounds of Seattle Birds Classification

Introduction

Bird species identification, utilizing bird features, is a task that traditionally would require expert knowledge and a substantial amount of time in the field. With modern technology such as scale audio recording and advancements in mathematics and the data field, it is now possible to automate this process using machine learning and the distinct signature of each bird's call. This project investigates whether neural networks can accurately identify bird species common in Seattle area based on their chirps.

The data used in this project comes from the Xeno Canto bird sounds archive, a crowd sourced collection of bird recordings from around the world[2]. From this collection, 12 Seattle area species were selected: American Crow (amecro), American Robin (amerob), Bewick's Wren (bewwre), Black capped Chickadee (bkcchi), Dark eyed Junco (daejun), House Finch (houfin), House Sparrow (houspa), Northern Flicker (norfli), Red winged Blackbird (rewbla), Song Sparrow (sonspa), Spotted Towhee (spotow), and White crowned Sparrow (whcspa). The raw audio clips were preprocessed into mel spectrograms, which are image like representations of sound that show how the frequency content of a signal changes over time. Each spectrogram has dimensions of 128 by 517, representing 128 frequency bins over a 3 second audio window. The dataset contains between 37 and 630 samples per species, reflecting a significant class imbalance that is addressed later in the report.

Three types of neural network models are built and compared in this project. First a binary classifier is trained on two species to validate the pipeline before scaling up. Then a 12-class model is built to classify all species simultaneously. Finally, the best performing model is applied to three unlabeled mystery clips to test how well it generalizes to new real-world audio. The models explored include feedforward neural networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs) with long short-term memory (LSTM) layers. The central question this project investigates is which network architecture performs best for bird call classification and whether balancing the training data improves results.

Theoretical Background

Neural Networks

A neural network takes an input vector $X = (X_1, X_2, \dots, X_p)$and builds a nonlinear function $f(X)$to predict a response Y [1, Ch. 10]. The basic building block is the hidden layer, where each unit computes an activation $A_k = g\left(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j\right)$, where $g(z)$ is a nonlinear activation function. The modern preferred choice is the ReLU (Rectified Linear Unit) function defined as $g(z) = max(0, z)$ which is computationally efficient and works well in practice. The activations from the hidden layer then feed into the output layer to produce the final prediction.

In a multilayer network activations from one hidden layer feed into the next, allowing the network to build up increasingly complex transformations of the input. For multiclass classification the output layer uses a softmax activation to convert raw scores into class probabilities. The network is trained by minimizing the cross-entropy loss using stochastic gradient descent. To prevent overfitting, dropout regularization randomly sets a fraction of activations to zero during training, which forces the network to learn more robust features [1, Sec. 10.7].

Convolutional Neural Networks

Convolutional neural networks are designed for image like data where spatial structure matters. Instead of connecting every input to every hidden unit, a CNN applies convolution filters that scan across the input looking for local patterns. A convolution filter is a small array of weights that gets applied to every patch of the input. If a patch resembles the filter, it produces a large value, otherwise a small one [1, Sec. 10.3].

After each convolution layer a pooling layer reduces the spatial dimensions by summarizing small blocks of the feature map. Max pooling takes the largest value in each block, which makes the representation more compact and provides some location invariance meaning small shifts in the input do not drastically change the output. A typical CNN stacks multiple convolution and pooling layers with the filter count increasing in deeper layers so the network can learn more complex features as it gets deeper. The final layers are fully connected dense layers that combine the learned features to make a classification decision. CNNs are well suited for spectrogram data because spectrograms are essentially images where local patterns in frequency and time are what distinguish one bird species from another.

Recurrent Neural Networks

Recurrent neural networks are designed for sequential data where the order of inputs matters. Unlike feedforward networks that process each input independently, an RNN maintains a hidden state that accumulates information from previous inputs in the sequence. At each time step the hidden state is computed from both the current input and the previous hidden state [1, Sec. 10.5].

The LSTM (Long Short Term Memory) is a popular variant of the RNN that uses gating mechanisms to control what information is remembered or forgotten over long sequences. This makes LSTMs better at capturing long range dependencies than simple RNNs. In our case spectrograms are treated as sequences of frequency bins over time, though in practice the CNN outperformed the RNN since the spatial structure of the spectrogram is more informative than treating it as a pure sequence.

Model Evaluation

Models are evaluated on a held-out test set that was not used during training. The primary metric is classification accuracy, which is the fraction of test samples correctly classified. However, accuracy alone is insufficient under class imbalance, so a per-species classification report is also produced, reporting precision, recall, and F1-score for each species. Precision measures how often a predicted label is correct; recall measures how many true samples of a species are correctly identified; and F1-score is their harmonic mean. Macro-averaged F1 weights all species equally regardless of sample count, making it the most informative single metric when class sizes differ substantially. Top-3 accuracy is also reported for multiclass models, indicating whether the correct species appears among the model's top three predictions. For multiclass problems a confusion matrix is used to show which species are most frequently confused with each other. Early stopping halts training once validation loss stops improving, preventing overfitting without needing to fix the number of epochs in advance.

Methodology

The data was provided as an HDF5 file containing mel spectrograms for 12 bird species with dimensions of 128 by 517 per sample. Each species array was transposed and reshaped to match the input format required by Keras, and labels were one hot encoded. The dataset was split 80/20 into training and test sets with stratification. Due to class imbalance ranging from 37 to 630 samples per species, a second downsampled dataset was created by capping each species at 100 samples for comparison. For the three external test clips, the same preprocessing pipeline was applied using librosa with a hop length of 128 and window length of 512 to match the original spectrograms.

A binary classifier was first built on two species (Dark-eyed Junco and House Finch) to validate the pipeline, comparing a CNN and an RNN architecture. The full 12-class problem was then tackled using three architectures: a feedforward neural network as a baseline, a CNN with three convolutional blocks, and an RNN with stacked LSTM layers. Since the CNN performed best, three CNN variants were compared on the downsampled dataset varying filter counts and dropout rate. All models were trained with categorical cross-entropy loss, early stopping on validation loss with patience of 5, and evaluated on the held-out test set using classification accuracy, Top-3 accuracy, per-species F1-score, and confusion matrices.

Results

For the binary classification, the CNN achieved a test accuracy of 90.48%, outperforming the RNN which achieved 85.71%. The confusion matrix for the CNN is shown in Figure 1. The model correctly classified the majority of both Dark-eyed Junco and House Finch samples with very few misclassifications between the two species. The classification report confirms high precision and recall for both species, validating the pipeline before scaling up.

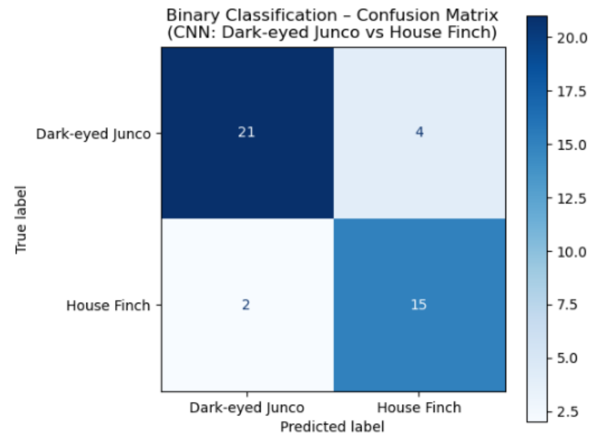

Figure 1: Binary classification confusion matrix

Scaling to all 12 species proved more challenging. The feedforward network achieved a test accuracy of 28.43% and the RNN achieved 31.74%, while the CNN again performed best at 50.88%. However, accuracy alone is misleading here. The confusion matrix in Figure 2 shows that House Sparrow dominates with 108 correct predictions and Song Sparrow with 34, inflating overall accuracy while masking poor performance elsewhere. House Finch achieves only 1 correct prediction, and both Black-capped Chickadee and Northern Flicker achieve zero. Bewick's Wren is split almost entirely between House Sparrow (7 samples) and Song Sparrow (11 samples), and Red-winged Blackbird loses 12 samples to House Sparrow and 10 to Song Sparrow, showing that the two most frequent classes absorb the majority of misclassifications across the board. The Top-3 accuracy is considerably higher than Top-1, indicating the correct species frequently appears among the model's top predictions even when not ranked first. The macro-averaged F1 is substantially lower than the weighted F1, which quantifies the class imbalance effect more precisely than accuracy alone.

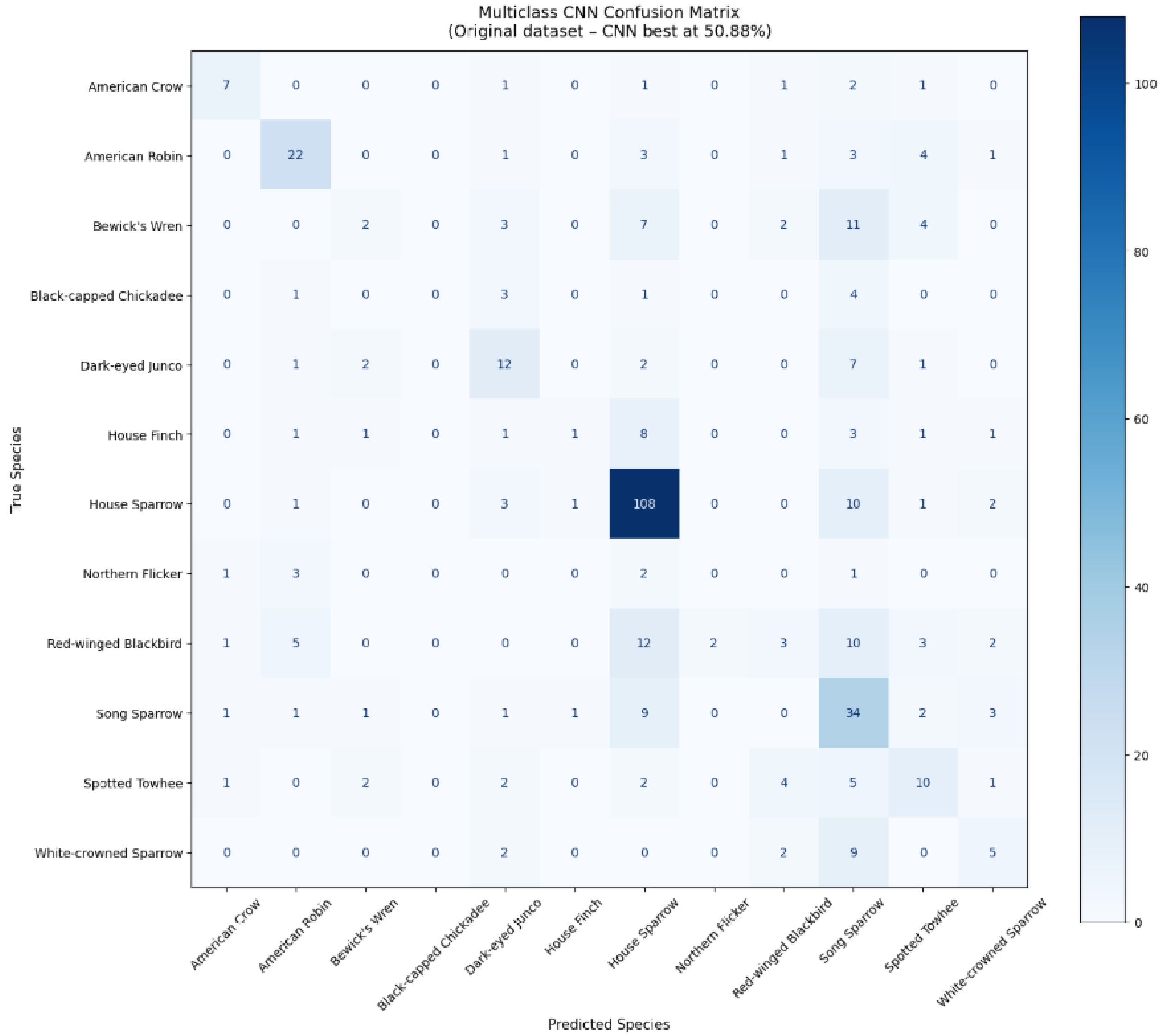

Figure 2: Multiclass classification confusion matrix

To investigate whether class imbalance was driving these results, a second set of models was trained on a downsampled dataset capping each species at 100 samples. Three CNN variants were tested and their validation curves are shown in Figure 3.

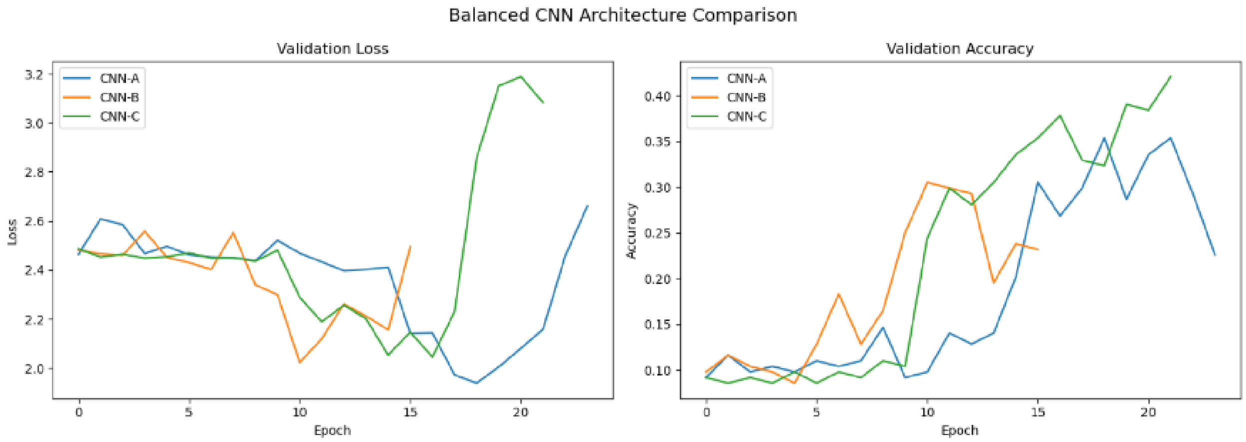

Figure 3: validation loss and validation accuracy

CNN-B with filter counts of 16, 32, and 64 and a dropout rate of 0.3 performed best at 34.15%, followed by CNN-C at 34.63% and CNN-A at 32.68%. The narrow margins between all three variants indicate that architecture differences have limited impact when training data is constrained to 100 samples per species. CNN-A's larger filters did not provide an advantage, suggesting features relevant to bird call classification at this dataset size are captured equally well by a simpler architecture. The confusion matrix for CNN-A on the downsampled dataset is shown in Figure 4. House Sparrow (13 correct), American Robin (12), and Dark-eyed Junco (10) are the strongest performers. Black-capped Chickadee and Northern Flicker again achieve zero correct predictions, confirming their difficulty is acoustic rather than purely a sample size issue. Red-winged Blackbird improves from 3 to 6 correct predictions compared to the original model. Despite lower overall accuracy, predictions are more evenly distributed across species and the macro-averaged F1 improves relative to the original model.

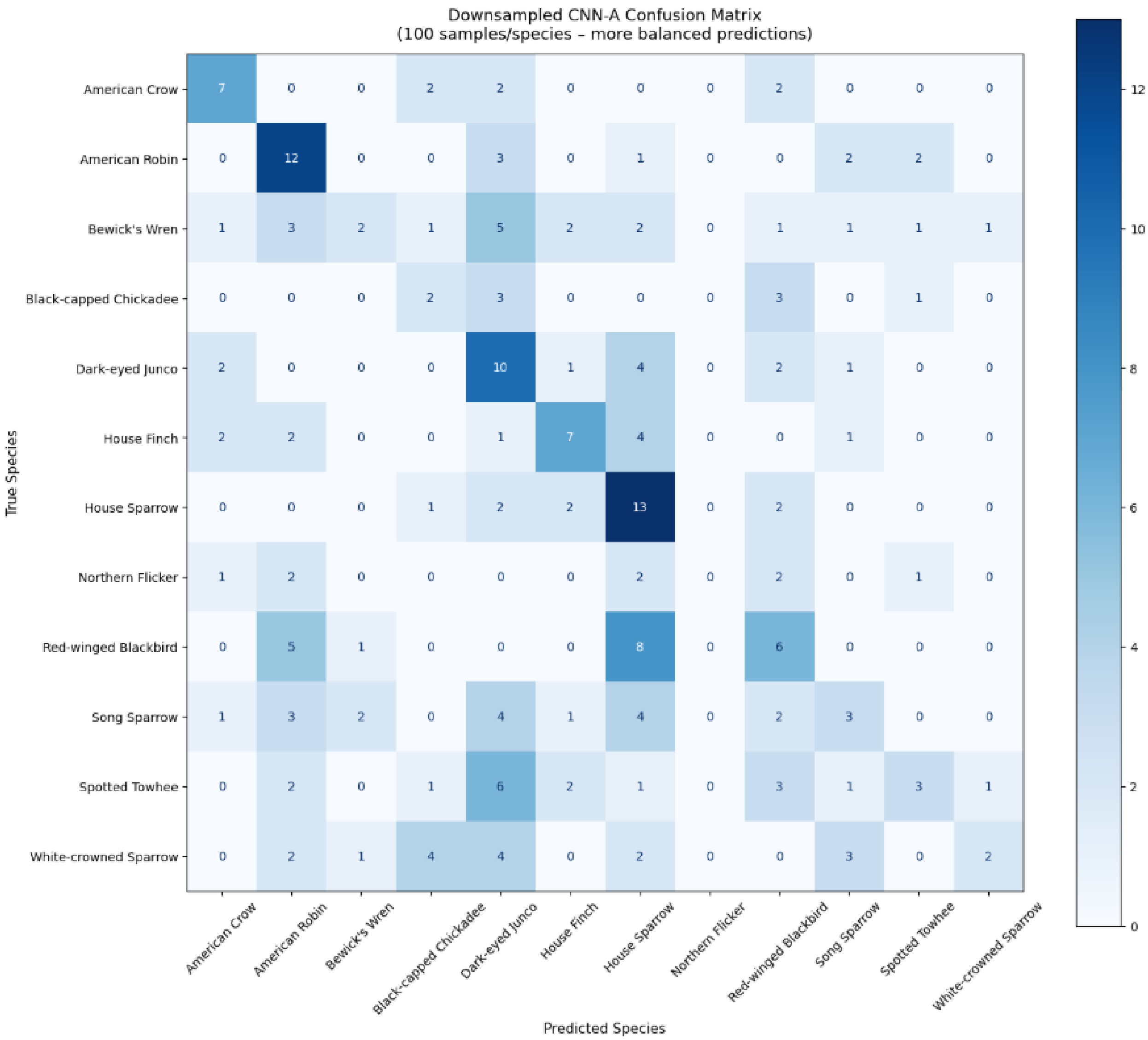

Figure 4: down sampled multiclass CNN confusion matrix

The spectrograms for the three external test clips are shown in Figure 5. The original model predicted White-crowned Sparrow for all three clips with 100% confidence, reflecting complete collapse to a dominant class. The downsampled CNN-B produced substantially more varied predictions as shown in Table 1. Test2 was predicted as Red-winged Blackbird at 35.4%, test1 as Black-capped Chickadee at 18.3%, and test3 as House Sparrow at 12.8%, the lowest confidence of the three, reflecting genuine uncertainty across the clips.

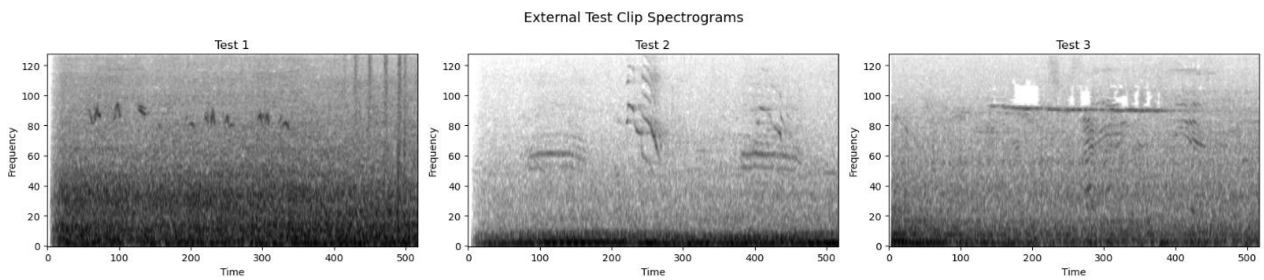

Figure 5: External test clips spectrograms

Clip	Original Prediction	Confidence	Downsampled Prediction	Confidence	Top 3 (Downsampled)
test1.mp3	White-crowned Sparrow	100%	Black-capped Chickadee	18.3%	Black-capped Chickadee, House Sparrow, House Finch
test2.mp3	White-crowned Sparrow	100%	Red-winged Blackbird	35.4%	Red-winged Blackbird, House Sparrow, American Robin
test3.mp3	White-crowned Sparrow	100%	House Sparrow	12.8%	House Sparrow, House Finch, Red-winged Blackbird

Table 1: Predictions for three external test clips from the original and downsampled CNN-B models

Discussion

The CNN outperformed both the feedforward network and the RNN across all experiments. The feedforward network struggled because flattening the spectrogram discards spatial structure, and the RNN performed only marginally better by treating frequency bins as a time sequence. The CNN's ability to detect local patterns in both frequency and time made it the most suitable architecture for this task. However, the original model's 50.88% accuracy is heavily inflated by House Sparrow (108 correct) and Song Sparrow (34 correct), while House Finch (1 correct), Red-winged Blackbird (3 correct), and Bewick's Wren (2 correct) are nearly invisible in its predictions. This makes accuracy an insufficient metric on its own, and motivates both the per-species classification report and the downsampled experiment.

The class imbalance had a clear and measurable effect on results. With House Sparrow contributing 630 samples compared to as few as 37 for Northern Flicker, the original model gravitates toward predicting majority classes as seen in the confusion matrix. House Sparrow and Song Sparrow act as catch-all prediction targets, absorbing misclassifications from nearly every other species. The macro-averaged F1, which weights all species equally, is substantially lower than the weighted F1, making this the clearest numerical signal of the imbalance problem. Downsampling to equal class sizes broke this concentration, and errors distributed more evenly. Red-winged Blackbird improved from 3 to 6 correct predictions. The trade-off is that overall accuracy dropped from 50.88% to 34.15%, but the model becomes meaningfully more useful for identifying the full range of species rather than defaulting to the two most common ones.

The confusion matrices also reveal a sparrow cluster — House Sparrow, Song Sparrow, White-crowned Sparrow, and Dark-eyed Junco — where species are mutually confused with each other across both experiments. This is consistent across both the original and downsampled settings, suggesting it is a genuine acoustic similarity problem that more data alone would not fully solve. Black-capped Chickadee and Northern Flicker achieve zero correct predictions in both experiments, confirming their failure is driven by acoustic overlap rather than sample count alone. The external test results reinforce the imbalance issue starkly. The original model predicted White-crowned Sparrow for all three clips with 100% confidence, with no discrimination between clips. The downsampled CNN-B predicted three different species across the three clips with confidence values ranging from 12.8% to 35.4%, which is far more consistent with the genuine difficulty of the task. Among the downsampled CNN variants, CNN-B's simpler architecture outperformed the larger CNN-A, reinforcing that model complexity should be matched to available training data size.

The main limitation is the small dataset. With some species having fewer than 40 samples, data augmentation such as time shifting or frequency masking could significantly improve performance. The macro vs. weighted F1 gap in the classification reports provides a concrete target for future improvement; closing that gap would indicate the model is treating all species more equitably. Training time was also a constraint given the large spectrogram input size of 128 by 517.

Conclusion

This project demonstrated that convolutional neural networks can classify bird species from spectrogram data with reasonable accuracy, achieving 50.88% on 12 species using the original dataset. However, this accuracy figure is misleading, as per-species classification reports reveal that House Sparrow and Song Sparrow drive most correct predictions while several species achieve near-zero F1-scores. The CNN consistently outperformed both the feedforward and RNN architectures, and the results highlight that class imbalance is a significant challenge when working with real-world wildlife audio data where some species are naturally more recorded than others. Among the downsampled CNN variants, the simpler CNN-B outperformed the larger CNN-A, reinforcing that model complexity should be calibrated to available training data.

The findings have practical implications for automated bird monitoring. A model that can reliably identify species from short audio clips could support conservation efforts by tracking species presence across large areas without requiring human experts in the field. However the results also show that model performance is heavily dependent on having sufficient and representative training data, which is a real constraint when working with rare or hard to record species.

Future work could improve on these results through data augmentation, larger training sets, or pretrained audio models that have already learned general acoustic features. Closing the gap between macro and weighted F1 would be a concrete measurable goal for future iterations. The external test predictions also suggest that detecting multiple simultaneous bird calls remains an open challenge that would require more specialized approaches such as multi-label classification.

References

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[2] W. Vellinga and R. Planqué, “The Xeno-canto collection and its relation to sound recognition and classification,” in *Proc. CLEF 2015 Working Notes*, 2015. [Online]. Available: https://xeno-canto.orghttps://xeno-canto.org